显示测试（FramBuffer）

命令行测试

确定设备节点

```
ls /dev/fb*
cat /proc/fb
```

花屏测试

```
cat /dev/urandom > /dev/fb0

# 如果没有视频中的现象，请断电之后，再重新上电尝试
# 如果仍然无现象，请尝试下面的“colorbar测试”，有现象即可
```

colorbar测试

```
cat /dev/zero > /dev/fb0
echo 8 > /sys/class/disp/disp/attr/colorbar
```

编码测试

framebuffer_test.c（lvgl_100ask_course_v9\part3\02_codes\lesson_2-2\framebuffer_test.c）

```
#include <sys/mman.h>
#include <sys/types.h>
#include <sys/stat.h>
#include <unistd.h>
#include <linux/fb.h>
#include <fcntl.h>
#include <stdio.h>
#include <string.h>
#include <sys/ioctl.h>
#include <time.h>

static int fd_fb;
static struct fb_fix_screeninfo fix;    /* Current fix */
static struct fb_var_screeninfo var;    /* Current var */
static int screen_size;
static unsigned char *fb_base;
static unsigned int line_width;
static unsigned int pixel_width;

void lcd_put_pixel(void *fb_base, int x, int y, unsigned int color)
{
    unsigned char *pen_8 = fb_base+y*line_width+x*pixel_width;
    unsigned short *pen_16;
    unsigned int *pen_32;
```

```
    unsigned int red, green, blue;

    pen_16 = (unsigned short *)pen_8;
    pen_32 = (unsigned int *)pen_8;

    switch (var.bits_per_pixel)
    {
        case 8:
        {
            *pen_8 = color;
            break;
        }
        case 16:
        {
            /* 565 */
            red   = (color >> 16) & 0xff;
            green = (color >> 8) & 0xff;
            blue  = (color >> 0) & 0xff;
            color = ((red >> 3) << 11) | ((green >> 2) << 5) | (blue >> 3);
            *pen_16 = color;
            break;
        }
        case 32:
        {
            *pen_32 = color;
            break;
        }
        default:
        {
            printf("can't surport %dbpp\n", var.bits_per_pixel);
            break;
        }
    }
}

void lcd_draw_screen(void *fb_base, unsigned int color)
{
    int x, y;
    for (x = 0; x < var.xres; x++)
        for (y = 0; y < var.yres; y++)
            lcd_put_pixel(fb_base, x, y, color);
}

int main(int argc, char **argv)
{
    int i;
    int ret;
    int nBuffers;
    unsigned int colors[] = {0xFFFF0000, 0xFF00FF00, 0xFF0000FF, 0, 0xFFFFFFFF};  /*
0x00RRGGBB */
    struct timespec time;

    time.tv_sec  = 0;
    time.tv_nsec = 100000000;

    fd_fb = open("/dev/fb0", O_RDWR);
```

```
    if (fd_fb < 0)
    {
        printf("can't open /dev/fb0\n");
        return -1;
    }

    if (ioctl(fd_fb, FBIOGET_FSCREENINFO, &fix))
    {
        printf("can't get fix\n");
        return -1;
    }

    if (ioctl(fd_fb, FBIOGET_VSCREENINFO, &var))
    {
        printf("can't get var\n");
        return -1;
    }

    line_width  = var.xres * var.bits_per_pixel / 8;
    pixel_width = var.bits_per_pixel / 8;
    screen_size = var.xres * var.yres * var.bits_per_pixel / 8;

    nBuffers = fix.smem_len / screen_size;
    printf("nBuffers = %d\n", nBuffers);

    fb_base = (unsigned char *)mmap(NULL , fix.smem_len, PROT_READ | PROT_WRITE, MAP_SHARED,
fd_fb, 0);
    if (fb_base == (unsigned char *)-1)
    {
        printf("can't mmap\n");
        return -1;
    }

    while (1)
    {
        /* use single buffer */
        for (i = 0; i < sizeof(colors)/sizeof(colors[0]); i++)
        {
            lcd_draw_screen(fb_base, colors[i]);
            nanosleep(&time, NULL);
        }
    }

    munmap(fb_base , screen_size);
    close(fd_fb);

    return 0;
}
```

编译:

```
export PATH=$PATH:/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/prebuilt/rootfsbuilt/arm/toolchain-sunxi-glibc-gcc-830/toolchain/bin
export CROSS_COMPILE=arm-openwrt-linux-gnueabi
export STAGING_DIR=/home/ubuntu/100ask/T113s4_Lvgl9-Class_TinaSDK-V1/out/t113_s4/100ask_lvgl9/openwrt/staging_dir/target

arm-openwrt-linux-gnueabi-gcc framebuffer_test.c -o framebuffer_test
```

上传到开发板：

```
# 在ubuntu连接开发板的adb
adb push ./framebuffer_test /mnt/UDISK
```

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- 社区交流： https://forums.100ask.net
- QQ技术交流群（如群满，请加qq：401684796 验证备注：LVGL）：962138234
- 微信交流群：添加微信： baiwenkeji_fae 验证备注：LVGL